

Executive Summary –

Telco - Customer Churn Analysis

1. Project Overview

This project analyzes customer churn to understand **why customers leave the company and which customer characteristics, services, contracts, and payment methods are associated with higher churn.**

The analysis was performed using **Python, Pandas, NumPy, Matplotlib, and Seaborn**. The dataset contains **7,043 customers and 21 variables**, covering customer demographics, tenure, subscribed services, contract type, billing preferences, payment method, monthly charges, total charges, and churn status.

The analysis followed an exploratory data analysis approach:

- Data loading and inspection
 - Data type and structure validation
 - Missing-value checking
 - Duplicate-value checking
 - Data-type conversion for **TotalCharges**
 - Customer churn distribution analysis
 - Demographic analysis
 - Senior Citizen analysis
 - Tenure analysis
 - Contract analysis
 - Service-level analysis
 - Payment-method analysis
 - Business interpretation of the observed patterns
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2. Data Quality and Preparation

The dataset contains **7,043 records across 21 columns**.

The analysis identified:

- **7,043 customers**
- **21 variables**
- **0 missing values** after the data-cleaning process
- **0 duplicate rows**
- **0 duplicate customer IDs**

The **TotalCharges** column was initially stored as an object/string field. Blank values were replaced with 0 where the customer's tenure was 0, and the column was converted to a numerical data type.

This preparation made the dataset suitable for further statistical analysis and visualization.

Dataset Characteristics

Metric	Value
Total customers	7,043
Total variables	21
Missing values	0
Duplicate records	0
Duplicate customer IDs	0
Average tenure	32.37 months
Median tenure	29 months
Maximum tenure	72 months
Average monthly charge	64.76
Average total charge	2,279.73
Senior Citizen customers	16.21%

The average customer has been with the company for approximately **32 months**, while the median tenure is **29 months**.

3. Overall Customer Churn

The overall churn analysis shows that **26.54% of customers have churned**, while approximately **73.46% have remained with the company**.

In terms of customer count:

- **1,869 customers churned**
- **5,174 customers did not churn**

This means that roughly **1 out of every 4 customers** in the dataset has churned.

Business Interpretation

A churn rate of **26.54%** indicates that customer retention is an important business area to investigate. The subsequent analysis therefore focuses on identifying customer characteristics and service patterns associated with churn.

The analysis does **not establish that any individual factor causes churn**; rather, the charts show associations between customer characteristics and churn outcomes.

4. Churn by Gender

The notebook includes a comparison of churn across male and female customers.

The gender analysis does not show a major difference in churn rates between the two groups. Based on the dataset:

- Female customers: approximately **26.9% churn**
- Male customers: approximately **26.2% churn**

The difference is relatively small compared with the larger differences observed for factors such as contract type, tenure, and payment method.

Business Interpretation

Gender does not appear to be one of the strongest differentiating variables in this exploratory analysis. Therefore, the more substantial churn patterns are found in **customer tenure, contract structure, services, and payment methods**.

5. Churn by Senior Citizen Status

The notebook specifically investigates churn among Senior Citizen customers using a **100% stacked bar chart**.

Senior Citizens represent approximately **16.21% of the overall customer base**, while approximately **83.79% are non-senior customers**.

The churn comparison shows a noticeable difference:

Customer Group	Churn Rate
Senior Citizen	≈ 41.68%
Non-Senior Citizen	≈ 23.60%
Overall	26.54%

Therefore, the churn rate among Senior Citizens is substantially higher than the overall customer churn rate.

Business Interpretation

The chart indicates that **Senior Citizen customers form a smaller portion of the customer base but have a comparatively higher churn rate**.

This makes the Senior Citizen segment an important population for further investigation. Additional analysis could examine whether their higher churn is connected to factors such as contract type, tenure, service usage, payment method, or monthly charges.

6. Customer Tenure and Churn

The tenure analysis is one of the important findings in the notebook.

The histogram comparing tenure with churn shows a clear pattern: **customers with shorter tenure are more frequently associated with churn**, while customers who have remained with the company for longer periods tend to be more stable.

The dataset has:

- Minimum tenure: **0 months**
- Median tenure: **29 months**
- Average tenure: **32.37 months**
- Maximum tenure: **72 months**

The notebook specifically notes that customers who have used the company's services for a longer period tend to stay, while customers with approximately **1–2 months of tenure show higher churn**.

Business Interpretation

This suggests that the **early customer lifecycle is an important retention period**.

New customers may require additional attention during onboarding because they have not yet developed a long-term relationship with the company.

A practical business response could therefore include:

- Improved onboarding
 - Early customer satisfaction checks
 - First-month and second-month engagement campaigns
 - Faster resolution of early service problems
 - Early identification of customers showing signs of dissatisfaction
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7. Contract Type and Churn

The contract analysis shows one of the strongest differences in the dataset.

The churn rates by contract type are approximately:

Contract Type	Churn Rate
Month-to-month	42.71%
One year	11.28%
Two year	2.83%
Overall	26.54%

The chart demonstrates that customers with **month-to-month contracts have substantially higher churn** than customers with longer-term contracts.

Approximately **42.7% of month-to-month customers churned**, compared with approximately **11.3% of one-year customers** and only **2.8% of two-year customers**.

Business Interpretation

Contract type represents one of the clearest associations with churn in this analysis.

The data suggests that customers without a long-term contractual commitment are considerably more likely to leave.

However, this should be interpreted as an **association rather than proof that contract type itself causes churn**. Other characteristics may also influence this relationship.

From a retention perspective, the business could investigate whether suitable incentives, service benefits, or customer education could encourage customers to consider longer-term contracts.

8. Service Usage Analysis

The notebook includes multiple service-level charts covering:

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

The service analysis shows that customers who remain with the company frequently use services such as **Phone Service, Internet Service, Online Security, Online Backup, Tech Support, and Streaming services**.

The notebook's interpretation also identifies that **churn rates are noticeably higher when certain additional services are not used or are unavailable**.

Internet Service

One particularly noticeable difference is associated with Internet Service.

Approximate churn rates in the dataset are:

Internet Service	Churn Rate
DSL	≈19.0%

Fiber optic **≈41.9%**

No Internet service **≈7.4%**

The Fiber optic segment therefore shows a substantially higher churn percentage than the DSL segment.

Business Interpretation

This finding should not automatically be interpreted as "Fiber optic causes churn." Instead, it indicates that customers in the Fiber optic segment have a higher observed churn rate.

Further investigation could examine whether this relationship is connected with:

- Higher monthly charges
 - Contract type
 - Lack of additional support/security services
 - Customer tenure
 - Payment method
 - Service experience
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9. Online Security and Other Support Services

The analysis also examines additional services such as:

- Online Security
- Online Backup
- Device Protection
- Tech Support

The charts indicate that customers using certain additional services tend to show different churn patterns from customers who do not use those services.

For example, the Online Security comparison shows a substantially higher churn rate among customers without the service compared with customers who have the service.

Business Interpretation

Additional services may be associated with stronger customer engagement and perceived value.

The analysis therefore suggests an opportunity to investigate whether customers without security, backup, protection, or technical-support services require additional engagement or whether service adoption itself is related to longer customer retention.

Again, the analysis demonstrates **association rather than causation**.

10. Payment Method and Churn

The payment-method analysis is another important part of the project.

The notebook's chart compares churned and non-churned customers across:

- Electronic check
- Mailed check
- Bank transfer (automatic)
- Credit card (automatic)

The observed churn rates are approximately:

Payment Method	Churn Rate
Electronic check	45.29%
Mailed check	19.11%
Bank transfer (automatic)	16.71%
Credit card (automatic)	15.24%
Overall	26.54%

The **Electronic check** category stands out clearly.

Approximately **45.3% of customers using electronic check churned**, which is considerably higher than the overall churn rate of **26.54%**.

Business Interpretation

The payment-method chart suggests a strong association between **electronic-check payment and customer churn**.

The notebook specifically concludes that customers using electronic check are more likely to churn.

A business team could investigate whether this pattern is related to:

- Payment experience
- Billing friction
- Customer preferences
- Contract type
- Customer tenure
- Monthly charges
- Other characteristics of electronic-check users

The payment method itself should not automatically be treated as the root cause without additional analysis.

11. Major Patterns Identified

When the different charts are considered together, several important patterns emerge.

1. Overall churn is significant

26.54% of customers churned, meaning more than one-quarter of the customer base in the dataset left the company.

2. Early-tenure customers require attention

Customers with very short tenure, particularly around the first few months, show higher churn in the tenure analysis.

3. Month-to-month contracts have substantially higher churn

The observed churn rate is approximately:

42.71% for month-to-month customers

compared with:

11.28% for one-year customers

and:

2.83% for two-year customers.

4. Senior Citizens show higher churn

The Senior Citizen segment has an observed churn rate of approximately **41.68%**, compared with approximately **23.60% among non-senior customers.**

5. Electronic-check customers show high churn

Approximately **45.29% of electronic-check customers churned**, making this payment category an important segment for further investigation.

6. Service usage is associated with different churn patterns

Customers using additional services such as security, backup, device protection, and technical support show different churn patterns from customers without these services.

7. Fiber-optic customers show higher observed churn

The Fiber optic group has an observed churn rate of approximately **41.9%**, compared with approximately **19.0% for DSL customers.**

12. Key Business Areas for Further Investigation

Based on the exploratory findings, the analysis identifies several areas that could be examined further:

Customer Onboarding

Because early-tenure customers show higher churn, the company could analyze the first few months of the customer journey to identify possible service or onboarding issues.

Contract Strategy

The large difference between month-to-month and longer-term contracts suggests that contract structure deserves further analysis.

Payment Experience

The high churn rate among electronic-check customers suggests that the billing and payment experience should be examined more closely.

Senior Citizen Segment

The higher churn rate among Senior Citizens indicates a need for deeper segmentation to understand the underlying customer experience.

Service Adoption

The relationship between additional services and churn could be analyzed to determine whether customers with security, backup, device protection, and technical support have different retention patterns.

Internet Service

The higher observed churn within the Fiber optic segment should be investigated alongside pricing, contract type, tenure, and support services.

13. Overall Executive Conclusion

The customer churn analysis provides a detailed view of the factors associated with customer retention and attrition across **7,043 customers and 21 variables**.

The overall churn rate is **26.54%**, meaning approximately one in four customers has churned. The strongest patterns identified in the exploratory analysis are associated with **customer tenure, contract type, Senior Citizen status, payment method, Internet Service, and the availability or usage of additional support and security services**.

The most noticeable segment-level differences include:

- **42.71% churn** among month-to-month contract customers
- **11.28% churn** among one-year contract customers
- **2.83% churn** among two-year contract customers
- **41.68% churn** among Senior Citizen customers
- **23.60% churn** among non-Senior Citizen customers
- **45.29% churn** among electronic-check users
- **41.9% churn** among Fiber optic customers
- **19.0% churn** among DSL customers
- **26.54% overall churn**

Overall, the analysis demonstrates that customer churn is not evenly distributed across the customer base. Certain customer segments show substantially different churn rates, making **segmentation-based retention analysis** more useful than treating all customers equally.

The project demonstrates the use of **Python-based exploratory data analysis, data cleaning, statistical summaries, categorical analysis, percentage-based comparisons, and visualization** to convert raw customer data into meaningful business insights.

The next analytical step would be to move beyond descriptive analysis and build a **customer churn prediction model** or a **Power BI churn dashboard**, where customers can be segmented by churn risk, contract, tenure, services, payment method, and customer characteristics.